

JEE MAIN REVISION MASTER: SETS AND RELATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Sets, Power Sets & Operations (De Morgan's Laws)	15 PYQs	Set Theory
High	Relations: Reflexive, Symmetric, Transitive & Equivalence	8 PYQs	Relations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Sets, Power Sets & Operations (De Morgan's Laws)	$n(P(A)) = 2^n$, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, $(A \cup B)' = A' \cap B'$
Relations: Reflexive, Symmetric, Transitive & Equivalence	$\text{Total Relations on Set } A = 2^{n^2}$, $\text{Reflexive Relations} = 2^{n^2 - n}$, $\text{Symmetric Relations} = 2^{\frac{n(n+1)}{2}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Sets, Power Sets & Operations (De Morgan's Laws) (Very High Yield — 15 PYQs)

Core Principle: Set theory principles: Number of subsets of a set with n elements is 2^n ; number of proper subsets is $2^n - 1$.
1. Cardinality formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

■ **Caution / JEE Pitfall:** Power set of an empty set $\emptyset$ is NOT empty; it contains one element: $P(\emptyset) = \{\emptyset\}$, so $n(P(\emptyset)) = 2^0 = 1$!

■ **Speed Trick:** Number of subsets with at least one element is $2^n - 1$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If $n(A)=2$ and the number of relations from set A to set B is 1024, then $n(B)$ is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "If $A=1,2,3,4,\dots,10$, then the number of non-empty subsets of A containing only even number is..." [Ans: A]
- [2026 Shift] Q#4 (MCQ): "If $A=a,b,c,d,e,f$, then the number of subsets of A which contains at least 2 elements is..." [Ans: C]

3.2 Relations: Reflexive, Symmetric, Transitive & Equivalence (High Yield — 8 PYQs)

Core Principle: Types of relations on set A with n elements: Total relations = 2^{n^2} . Reflexive: $(a, a) \in R$ for all $a \in A$. Symmetric: $(a, b) \in R$ implies $(b, a) \in R$. Transitive: $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$. Equivalence relation: satisfies reflexive, symmetric, and transitive simultaneously.

■ **Caution / JEE Pitfall:** Empty relation on a non-empty set is symmetric and transitive, but NOT reflexive!

■ **Speed Trick:** The relation 'is parallel to' among lines is an equivalence relation; 'is perpendicular to' is symmetric but NOT reflexive or transitive.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Let R be the relation in the set ■ given by $R = \{(a, b) : a = b - 2, b > 6\}$. Which of the following is the correct answer?..." **[Ans: N/A]**
- **[2025 Shift] Q#3 (MCQ):** "Let $A = \{a, b, c\}$, then the number of equivalence relations on A containing (b, c) is..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "Consider the following statements : Statement(I) : The set of all solutions of the linear inequalities $3x + 8 < 17$ and $2x + 8 \geq 12$ are $x < 3$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Let R be the relation in the set ■ given by $R = \{(a, b) : a = b - 2, b > 6\}$. Which of the following is the correct answer?

- (A) $(2, 4) \in R$
- (B) $(3, 8) \in R$
- (C) $(6, 8) \in R$
- (D) $(8, 7) \in R$

Correct Answer: None

Solution Analysis: For a pair (a, b) to be in the relation R, it must satisfy both conditions: $a = b - 2$ and $b > 6$. Check the options: For $(2, 4)$: $2 = 4 - 2$ This is true, but $4 > 6$ is false. So $(2, 4) \notin R$. For $(3, 8)$: Here $b = 8$, so $b > 6$ is true. But $b - 2 = 8 - 2 = 6$ so a should be 6, not 3. Hence $(3, 8) \notin R$. For $(6, 8)$: Here $b = 8$, so $b > 6$ is true. Also, $b - 2 = 8 - 2 = 6$ So $a = 6$. Both conditions are satisfied. Thus, $(6, 8) \in R$ So the correct answer is: Option C

Benchmark Problem 2: KCET 2026 (Q#2)

If $n(A) = 2$ and the number of relations from set A to set B is 1024, then $n(B)$ is

- (A) 2
- (B) 5 **[Correct]**
- (C) 5^2
- (D) 2^5

Correct Answer: B

Solution Analysis: The number of relations from set A to set B is given by $2^{n(A) \times n(B)}$. Here, $n(A) = 2$ and the number of relations is 1024. So, $2^2 \times n(B) = 1024$. Now, $1024 = 2^{10}$. So, $2^{2n(B)} = 2^{10}$. Since the bases are the same, $2n(B) = 10$, $n(B) = 5$. So, the correct option is 5 Option B.